

Analysis of Scientific Data



Lecture 5

Sampling Distributions

Binomial Distribution

A *Bernoulli trial* is a random process which has two possible outcomes, usually labelled “success” and “failure”.

The count of total successes in a series of **independent** Bernoulli trials, with a **constant probability of success** p , has what is called the *Binomial distribution*.

If X is the total count from n independent trials, each with success probability p , then we write $X \sim \text{Binomial}(n, p)$.

Poll Question

Suppose proportion p of all students live at home. If we take a random sample of 5 students, what is the probability that exactly 5 of them live at home?

1. $5p$
2. p^5
3. $(1 - p)^5$
4. $5p^5$

Poll Question

Suppose proportion p of all students live at home. If we take a random sample of 5 students, what is the probability that exactly 4 of them live at home?

1. p^4
2. $4p^5$
3. $4(1 - p)^5$
4. $5p^4(1 - p)$

Suppose proportion p of all students live at home. If we take a random sample of 5 students, what is the probability that at least 3 of them live at home?

What is the expected number of students living at home from a sample of 5 students?

What is the standard deviation of the number of students living at home from a sample of 5 students?

Binomial Distribution

Definition

A random variable X has the *Binomial distribution* with parameters n and $0 < p < 1$ if

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k},$$

for $k = 0, 1, \dots, n$. We write $X \sim \text{Binomial}(n, p)$.

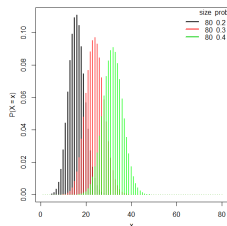
Properties

$$E(X) = np, \quad \text{var}(X) = np(1 - p).$$

Binomial probabilities with R

```
dbinom(x, size, prob)
```

```
pbinom(x, size, prob)
```



Normal Distribution

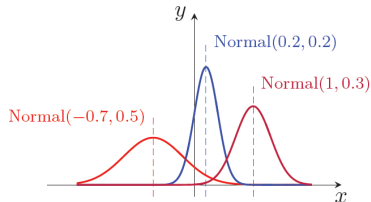
Definition

We say a continuous random variable X has a *Normal distribution* with parameters μ and $\sigma > 0$ if its density curve is of the form

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty.$$

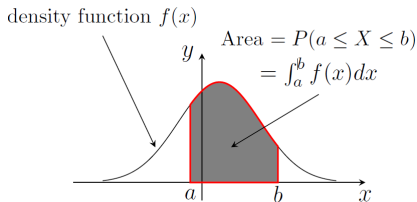
We write $X \sim \text{Normal}(\mu, \sigma)$.

What are $E(X)$ and $\text{sd}(X)$?



Normal probabilities

The probability that a normal random variable X takes on a value in a specific interval (a, b) is found by determining the area enclosed by the x -axis, the lines $X = a$ and $X = b$, and the bell shape curve.



Normal probabilities with R

```
dnorm(x, mean, sd)
```

```
pnorm(x, mean, sd)
```


Normal Probabilities

Suppose the heights of students have a Normal distribution with mean 172 cm and standard deviation 10 cm.

If we pick a student at random, what is the probability they are taller than 180 cm?

What is the probability they are between 170 cm and 180 cm tall?

Suppose $X \sim \text{Normal}(\mu, \sigma)$. Consider the new random variable

$$Z = \frac{X - \mu}{\sigma}.$$

What are $E(Z)$ and $\text{sd}(Z)$?

Poll Question

Suppose $Z \sim \text{Normal}(0, 1)$. What is the *median* of Z ?

1. 0.0000
2. 0.6745
3. 0.7071
4. 1.0000

Poll Question

Suppose $Z \sim \text{Normal}(0, 1)$. What is the *third quartile* of Z ?

1. 0.0000
2. 0.6745
3. 0.7071
4. 1.0000

1.5 $\times$ IQR Rule

Assuming a normal distribution, how often does the 1.5 $\times$ IQR rule for box plots make a 'mistake'?

Population

Suppose this semester's student survey represents a *population*.

This population's travel times have mean $\mu = \underline{\hspace{2cm}}$ mins and standard deviation $\sigma = \underline{\hspace{2cm}}$ mins.

What shape is the distribution of travel times?

1. Symmetric
2. Skewed to the right
3. Skewed to the left

Suppose we take a sample of $n = 4$ students from this population and record their travel times.

We can use the sample mean of these 4 times to estimate the population mean.

But different samples will give different estimates...

$$n = 4$$

Let's do this lots of times.

How can we describe the variability we see in the sample means?

$$n = 16$$

Suppose we take larger samples with $n = 16$.

How will this affect the distribution of the sample mean?

$$n = 64$$

Suppose we take even larger samples with $n = 64$.

How will this affect the distribution of the sample mean?

Distribution of the Sample Mean

Central Limit Theorem

Let $X_1, X_2, \dots, X_n$ be independent and identically distribution (i.i.d.) random variables with mean μ and variance σ^2 . Then the **sample mean** $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ has *approximately* the normal distribution

$$\bar{X} \underset{\sim}{\text{approx}} \text{Normal} \left(\mu, \frac{\sigma}{\sqrt{n}} \right).$$

Normal Approximation to the Binomial Distribution

For sufficiently large n , we have

$$\text{Binomial}(n, p) \approx \text{Normal} \left(\mu = np, \sigma = \sqrt{np(1-p)} \right).$$

